ROLE OF INSTITUTIONAL QUALITY IN HEALTH CARE FINANCING AND LIFE EXPECTANCY IN NIGERIA

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Abstract

This study examines the interactive role of institutional quality in the relationship between health expenditures (capital, recurrent, and out-of-pocket) and life expectancy in Nigeria from 1984 to 2023 using the ARDL approach. Findings indicate that capital health expenditure (CHE) significantly improves life expectancy when analyzed independently (coefficient = 2.647, p = 0.019), but this effect diminishes when institutional quality is introduced, highlighting governance deficiencies. Recurrent health expenditure (RHE) shows no significant impact, reflecting systemic inefficiencies, while out-of-pocket spending (OUE) fails to sustainably enhance life expectancy, exacerbating healthcare inequities. The study underscores that weak institutions undermine health financing effectiveness, necessitating governance reforms to optimize outcomes. Policy recommendations include strengthening institutional frameworks, reforming recurrent expenditure management, reducing reliance on OUE through expanded health insurance, and integrating governance reforms with health financing strategies. These measures are critical for Nigeria to achieve meaningful improvements in population health and life expectancy.

Keywords: Health expenditure, Institutional quality, Life expectancy, ARDL, Nigeria.

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1. Introduction

1.1 Background to the study

Effective health care funding has been a common strategy used by development-driven nations, organizations, leaders, and policymakers to improve the health of their citizens and life expectancy. Thus, expenditure on health is seen as a very important aspect of a person's wellbeing and one of the key ways of investing and ensuring improved life expectancy as well as human capital development. According to Edeme, Emecheta and Omeje (2017), improved health conditions have the potential to increase economic growth in both developed and developing nations by improving not only life expectancy but also human capital as a whole. Suffice to say, life expectancy and health spending are proportional. Health expenditure in Nigeria is categorized into public and private spending. Public health expenditure involves government funding of health services, infrastructure, and programs, while private health expenditure includes out-of-pocket payments made by individuals and households. According to the World Health Organization (WHO, 2020), over 70% of Nigeria's total health expenditure comes from out-of-pocket spending, one of the highest rates globally. This dependence on out-of-pocket payments limits access to healthcare for low-income populations, exacerbating inequality in healthcare delivery. The reliance on private expenditure reflects the government's limited commitment to health financing, with public health spending accounting for less than 5% of Nigeria's Gross Domestic Product (GDP), far below the 15% benchmark set by the Abuja Declaration of 2001.

Although, Nigeria has pledged to earmark 15 % of its annual budget to health sector but analysis of the federal ministry of health and its agencies between 2001 to 2021 revealed contrary. The highest the country has allocated to the Ministry is 6.08% in 2012. Thus, the average budgetary allocation to the Federal Ministry of Health (FMoH) in 20 years is 4.69%. This is below the 15% Abuja Declaration commitment, and a 10.31%. Notably, capital expenditure, which is expected to lead increased investment in the health sector, lags behind. For example, over a 20-year period, recurring costs account for 78% of overall health expenditure, while capital costs account for only 22% (Obi & Ozolua, 2021). The huge gap in favor of recurrent expenditure demonstrates the government's preference for Human Resources for Health and office overhead expenses. Inadequate government leadership, fragmented healthcare delivery, limited budget, subpar health infrastructure, imbalanced personnel distribution, and ineffective collaboration among key players hamper the health sector at large

(Adedokun, Adekanmbi & Uthman, 2017). Empirical evidence supports the assertion that institutional quality significantly influences the relationship between health expenditure and life expectancy. Rajkumar and Swaroop (2008) found that the effectiveness of public spending on health outcomes is closely linked to the quality of governance. In countries with strong institutions, health expenditure is more effective in improving health outcomes and increasing life expectancy. Conversely, in countries like Nigeria, where institutional quality is low, the impact of health expenditure remains marginal due to inefficiencies, corruption, and resource mismanagement. Addressing the issue of institutional quality is, therefore, critical to improving the dynamics of health care financing and life expectancy in Nigeria. Strengthening institutions requires measures to promote transparency, reduce corruption, and ensure accountability in health spending.

In conclusion, the interaction between health expenditure and life expectancy in Nigeria is heavily influenced by the country's low institutional quality. While increased health spending has the potential to improve health outcomes, the weak institutional framework undermines the effectiveness of such expenditures. Corruption, poor governance, and resource mismanagement continue to limit Nigeria's ability to achieve significant progress in life expectancy. This study seeks to explore the dynamics of health expenditure and life expectancy in Nigeria while highlighting the critical role of institutional quality in shaping these outcomes. By addressing institutional weaknesses, Nigeria can improve the efficiency of its health care financing, reduce mortality rates, and enhance life expectancy for its population.

1.2 Statement of the problem

Nigeria, the most populous country in Africa, has persistently struggled with poor health indicators despite its abundant natural resources and economic potential. According to the World Bank (2021), life expectancy in Nigeria was approximately 55 years, significantly lower than the global average of 73 years. This places Nigeria among the countries with the lowest life expectancy worldwide (Onwube et al., 2021). Contributing factors include inadequate health financing, corruption, inefficient resource allocation, and poor institutional frameworks that undermine health care delivery. These challenges highlight the critical role of institutional quality in influencing health expenditure outcomes and, consequently, life expectancy. Therefore, the role of institutional quality in determining the effectiveness of health care financing and its impact on life expectancy cannot be overstated, as weak institutions often hinder the allocation and utilization of resources in the health sector. The low institutional

quality in Nigeria further exacerbates the inefficiencies in health care financing. Institutions are pivotal in shaping governance, transparency, and accountability in public spending. In Nigeria, weak institutions often result in corruption, mismanagement of resources, and an inability to effectively implement health policies. For instance, Transparency International (2021) ranked Nigeria 154th out of 180 countries in its Corruption Perceptions Index, underscoring the severity of corruption in public sectors, including healthcare. Corruption diverts resources meant for health infrastructure, essential medical supplies, and healthcare personnel salaries, leaving hospitals underfunded, understaffed, and underequipped. This institutional weakness diminishes the impact of health expenditure on improving health outcomes and life expectancy.

Statistical trend further illustrate how institutional failures influence health outcomes in Nigeria. Despite an increase in health spending from 3.8% of GDP in 2005 to approximately 4.6% in 2020, the country continues to experience high mortality rates and poor health outcomes. According to the WHO (2020), Nigeria has one of the highest maternal mortality rates globally, with 917 deaths per 100,000 live births. Similarly, the under-five mortality rate remains alarmingly high at 114 deaths per 1,000 live births. These figures reflect not only inadequate health funding but also the mismanagement of available resources caused by poor governance and institutional inefficiency. Moreover, the inefficiency of institutions in Nigeria affects healthcare accessibility and quality, particularly in rural and underserved areas. Studies such as Akinyemi et al. (2019) demonstrate that disparities in health outcomes are more pronounced in regions with weaker institutional frameworks and governance. Rural areas, which constitute a significant portion of the population, suffer from a lack of health infrastructure, shortage of qualified medical professionals, and poor implementation of health programs. The inability of institutions to address these disparities exacerbates the already poor health indicators in Nigeria.

1.3 Justification for the study

The nexus between institutional quality, health care financing, and life expectancy has gained prominence in global development discourse, given its implications for achieving sustainable development goals (SDGs), particularly SDG 3, which focuses on ensuring healthy lives and promoting well-being for all. Despite substantial efforts to increase health care financing worldwide, disparities in health outcomes persist, particularly in low- and middle-income countries. This disparity underscores the critical role of institutional quality in shaping how

health care financing translates into improved health outcomes (Kim et al., 2018). Institutions, defined by their capacity to enforce policies, ensure accountability, and curb corruption, are fundamental for the effective allocation and utilization of health care resources. Weak institutions often lead to inefficiencies, resource mismanagement, and inequitable access to health care services, thereby undermining life expectancy gains (Acemoglu & Robinson, 2012). Conversely, countries with strong institutional frameworks are better equipped to transform financial investments in health into tangible outcomes, such as reduced mortality and increased life expectancy (Mackenbach, 2019).

Existing studies predominantly focus on the impact of health care financing or institutional quality on health outcomes in isolation. However, limited research examines their interplay and how institutional quality moderates the effectiveness of health care financing. This study fills this critical gap, offering insights into how governance reforms can enhance the efficiency of health financing policies. Furthermore, as global health systems face increasing challenges from pandemics, economic crises, and demographic shifts, understanding the institutional factors that improve health outcomes becomes even more vital (World Bank, 2021). By addressing these gaps, this research provides actionable insights for policymakers, enabling them to design context-specific strategies that strengthen institutional capacity, optimize health care spending, and improve population health outcomes.

1.4 Research Questions

In light of the problem identified, the following questions guide the course of this investigative endeavor. They include:

1. Does institutional quality significantly interact with capital health expenditure-life expectancy nexus?
2. What is the interactive role of institutional quality on recurrent health expenditure - life expectancy nexus?
3. How does institutional quality interactively impact out-of-pocket expenditure – life expectancy nexus?

1.5 Research Objectives

The over-arching objectives of the study is to investigate the role of institutional framework in the health care spending-life expectancy nexus. Hence, the following are specific objectives of the study which is to:

1. compute, using principal component analysis (PCA), a composite index for Nigeria institutional quality
2. examine the interactive role of institutional quality on capital health expenditure-life expectancy nexus
3. investigate the interactive role of institutional quality on recurrent health expenditure - life expectancy nexus
4. analyze the interactive role of institutional quality on out-of-pocket expenditure – life expectancy nexus.

1.6 Scope of the study

The study is focused on Nigeria and would employ a data scope spanning from 1984 to 2023. The justification for this choice of data scope is influenced by three factors. First of which includes data availability and the need to have equal start and end dates. Secondly, a data scope of 42 years proves appropriate to account for the conventional statistical degree of freedom of 30 (N-k) for time series analysis where N is the number of observation and k is the number of model (s) parameters (lags inclusive). Lastly, the period considered captures various seasons in which the country experience significant ups and lows in life expectancy and commitment to health as well as various forms of institutional reforms.

2. Literature Review

2.1 Theoretical Literature Review

A theoretical lens is essential for unpacking the intricate relationship between health care financing, life expectancy, and institutional quality. This sub-section critically reviews three relevant theories that underpin the study: the Grossman Theory of Health Capital, Wagner's Law of Increasing State Activities, and the Institutional Theory. These theories provide a robust foundation for understanding how health investments, public finance dynamics, and governance structures influence health outcomes in Nigeria.

2.1.1 Grossman Theory of Health Capital

Developed by Michael Grossman (1972), the Health Capital Model presents health as a durable capital stock that depreciates with age but can be increased through investment in healthcare and healthy behaviors. Grossman postulates that individuals derive both consumption and investment benefits from health. On one hand, good health increases utility; on the other, it

enhances productivity by reducing time lost to illness. Consequently, life expectancy can be improved through rational investment in health services, nutrition, education, and related resources. In Grossman's framework, healthcare expenditure—both public and private—is viewed as a form of investment in health capital. Therefore, increased health spending, particularly when efficient and equitably allocated, should lead to better health outcomes and longer life expectancy (Grossman, 1972). This theory aligns with studies like those by Kim and Lane (2013) and Boachie et al. (2018), which suggest a positive association between public health financing and life expectancy. However, it also implicitly assumes efficient health systems and governance, which may not always be the case in developing countries like Nigeria. In such contexts, institutional factors become critical intermediaries.

2.1.2 Wagner's Law of Increasing State Activities

Adolph Wagner's Law (1883) argues that as an economy develops, there is a natural and inevitable increase in public expenditure, including on health and welfare. According to Wagner, citizens' demand for healthcare and other public goods rises with income and modernization, prompting governments to expand their fiscal role. This theory justifies increased government health expenditure as a response to economic and demographic changes and implies that well-structured public financing should improve population health outcomes. In relation to life expectancy, Wagner's Law suggests that rising public health investment in growing economies like Nigeria should result in improved longevity. However, this outcome depends on the efficiency and governance of health spending. Empirical studies (e.g., Edeme et al., 2017; Arthur and Oaikhenan, 2017) reveal that despite increased public health expenditure in some Sub-Saharan African countries, health outcomes often remain suboptimal due to systemic inefficiencies and institutional weaknesses. Hence, the integration of institutional quality into the healthcare financing framework becomes essential for Wagner's proposition to hold in practice.

2.1.3 Institutional Theory

The Institutional Theory emphasizes the role of formal and informal institutions—rules, norms, and governance structures—in shaping social and economic outcomes. In health economics, this theory is particularly relevant in explaining how institutional quality, such as governance, corruption control, regulatory effectiveness, and political stability, affects the efficiency of health systems and the allocation of healthcare resources (North, 1990; Acemoglu & Robinson,

2012). Good institutions ensure transparency and accountability in public health financing, which translates into effective service delivery and improved health outcomes, including life expectancy. For instance, Socoliuc et al. (2022) and Koller et al. (2020) have demonstrated that institutional quality significantly moderates the relationship between health spending and health outcomes. Conversely, in settings where corruption is rampant or governance is weak, increased health spending may not translate into better health indicators (Vian, 2020; Achim et al., 2020). This perspective is particularly salient in the Nigerian context, where mismanagement of public resources has been a persistent concern

2.2 Methodological Literature Review

The purpose of this sub-section is to conduct a thorough analysis of the analytical approaches that were utilised in the study that was investigated. The advantages, disadvantages, and implications of these methodologies for the findings on the impact of healthcare financing and institutional quality on life expectancy are highlighted. The investigation is structured in a thematic manner in order to emphasise the variations and similarities in the methodologies used.

To begin with, mixed-effects regression models are utilised in panel data analysis by Kim and Lane (2013) and Edeme et al. (2017). These models contain both fixed and random effects during the analysis process. This methodology is able to successfully capture unobserved variation (for example, traits that are distinctive to a country), but it assumes linear linkages, which may mask nonlinear dynamics in the effects of healthcare expenditure. According to Boachie et al. (2018) and Rezapour et al. (2019), the generalised method of moments (GMM) and dynamic panel models are utilised in order to address the issue of endogeneity through the utilisation of lagged variables. Although generalised method of moments (GMM) improves causal inference, the fact that it is dependent on instrumental factors might result in weak-instrument bias if the exclusion requirements are not met.

Furthermore, both Zaman et al. (2017) and Arthur and Oaikhenan (2017) make use of static panel models (fixed/random effects) without incorporating endogeneity, which may help to shed light on their null results. Their methods are unable to differentiate between correlation and causation, which is a critical limitation due to the fact that decisions regarding health expenditures are made by the endogenous system. Meanwhile, Jaba et al. (2014) and Duba et al. (2018) use multivariate regression with insufficient robustness checks (for example, there

are no tests for autocorrelation or heteroskedasticity), which implies that the reliability of their positive correlations is called into question.

From the works of Socoliuc et al. (2022) and Rahman et al. (2022), structural equation modelling (SEM) was employed in order to determine the relationships that exist between institutional quality, governance, and health outcomes. Despite the fact that Structural Equation Modelling (SEM) is particularly useful for latent variables like "institutional quality," it requires rigorous theoretical assumptions on the paths that lead to causality. Vian (2020) and Koller et al. (2020) make use of qualitative case studies and policy syntheses, which, despite the fact that they lack generalisability, provide complex insights into the dynamics of corruption. The research conducted by Mackenbach and McKee (2015) and Makuta and Bernadette (2015) makes use of cross-country fixed-effects models that incorporate governance indices, such as those provided by the World Bank. Despite the fact that these studies take into consideration time-invariant confounding factors, they consider institutional quality to be universal and ignore sector-specific institutional dynamics (for example, corruption in the health sector as opposed to general governance).

2.3 Empirical Literature Review

The dynamics of interaction between health care financing and health outcomes is not new in the literature. Various empirical attempts have been made to understand this interaction using different methodological underpinnings thereby resulting in divergent findings and conclusions. With respect to the interactive mechanism of institutional framework, a growing body of knowledge is sprouting as well.

2.3.1 Health care financing - life expectancy nexus

Intuitively, it is expected that increased healthcare spending should translate into improved life expectancy at birth through a web of some complex trajectories that may guarantee an equitable access to healthcare services. However, findings from the literature are mixed and diffused with respect to the contributions of healthcare spending to life expectancy at birth. Some studies have indicated positive effects of health expenditure on life expectancy at birth (Kim and Lane 2013; Edeme et al. 2017; Boachie et al. 2018; Rezapour et al. 2019; Osakede 2020). For example, in a panel study using the mixed-effect model regression, Kim and Lane (2013) discovered that government health expenditure exerted a positive effect on life expectancy at birth. This finding was reaffirmed by Edeme et al. (2017) who conducted a study involving Nigeria. Similarly, a couple of different but related studies carried out in Ghana, middle and

high income countries have lent credence to the positive association between public health spending and life expectancy at birth (Boachie et al. 2018; Rezapour et al. 2019). In contrast, a study carried out in Bangladesh by Zaman et al. (2017) did not establish any significant relationship between health expenditure and life expectancy. Also, Arthur and Oaikhenan (2017) studied 40 Sub-Saharan African (SSA) countries and realised that government health spending did not have any significant positive effect on life expectancy at birth in the affected countries.

Furthermore, health economics and public health literature are replete with evidence noting out-of-pocket health expenditure usually dominating the health spending of most developing countries, including Nigeria (Novignon et al. 2012; Jaba et al. 2014; Eboh et al. 2016; Bein et al. 2017; Arthur and Oaikhenan 2017; Duba et al. 2018; Nketiah-Amponsah 2019). For instance, Jaba et al. (2014) discovered in their panel study involving 175 countries that private health expenditure had a significant positive effect on life expectancy. The same finding is also replicated by Arthur and Oaikhenan (2017) and Duba et al. (2018). On the other hand, some authors have reported no significant positive relationship between private health spending (out-of pocket expenditure) and life expectancy at birth (Sango-Coker and Bein 2018; Rezapour et al. 2019). It is obvious from the available literature that the findings emanating from the effects of health expenditure on life expectancy at birth were mixed. While some authors favoured positive effect of health expenditure on life expectancy, others noted otherwise. Also, methodologically, the literature reviewed for this study mostly utilised panel data at different periods.

2.3.2 Institutional quality - health outcomes nexus

Unfortunately, there is a limited body of research investigating the role of institutional quality and its sub-indices in the healthcare sector. For instance, Socoliuc et al. (2022) demonstrated that within the European Union, higher institutional quality is associated with reduced child mortality rates and improved life expectancy. Similarly, Rahman et al. (2022) indicated that good governance, economic growth, and renewable energy consumption contribute to improved life expectancy in ANZUS-BENELUX countries. Moreover, Vian (2020) emphasized that corruption undermines the capacity of health systems to promote better health, economic growth, and overall development. Therefore, addressing corruption through interventions and allocating resources towards prevention and control are crucial components of strengthening health systems for Universal Health Coverage (UHC). Additionally, Koller et

al. (2020) demonstrated that anti-corruption policies, transparency, and accountability lead to enhanced access to physicians, nurses, medical drugs, and the development of UHC. Furthermore, Achim et al. (2020) revealed that corruption has a negative impact on life expectancy and happiness. Mackenbach and McKee (2015) found a significant positive association between the quality of democracy, quality of government, and process and outcome indicators of healthy policies. Similarly, Makuta and Bernadette (2015) demonstrated that better governance quality is associated with improved health outcomes in Sub-Saharan Africa, indicating that public health expenditure is more effective in countries with higher levels of governance quality compared to those with lower levels.

Regrettably, there has been a lack of comprehensive research exploring the relationship between institutional quality and its impact on health outcomes, specifically life expectancy, resulting in significant knowledge gaps. Existing studies have primarily focused on examining the link between democracy indicators and health outcomes, such as infant mortality or life expectancy. However, the role of institutional quality has received limited attention. While studying the impact of democracy indicators on health system performance is valuable, it is essential to delve deeper and investigate the influence of the quality of institutions that underpin these democracy indicators. Therefore, the objective of this study is to bridge these gaps by examining the role of institutional quality in the dynamics of health care financing and life expectancy in Nigeria.

3. Research Methodology

3.1 Sources and Methods of data collection

In the course of this study, secondary data is sourced from World Development Indicators (WDI), Nigerian Bureau of Statistics (NBS), and World Governance indicator (WGI) spanning across 1980-2022. The dependent variables being life expectancy rate and independent variables (health care financing) is proxy as capital health expenditure, recurrent health expenditure and out-of-pocket expenditure. The interactive variable (institutional quality) will be measured by a composite index created from various governance indices which includes Control of Corruption, Voice and Accountability, Political Stability and Absence of Violence/Terrorism, Rule of Law, Regulatory Quality, and Government Effectiveness. The control variables for the study includes GDP per capita, and Population rate.

3.2 Theoretical Framework

Health is a key aspect of human capital, and it is considered a long-lasting factor that is positively associated with the economic productivity of labor. In the literature, the baseline theoretical framework for modeling health has been proposed by Grossman (1972). The proposed health production function is considered one of the key tools for analyzing a country's health state. This health production function is similar to the traditional production function, which shows the socioeconomic input and output association. The dependent variables of the health production function are measured with reduced morbidity or life expectancy; however, the inputs involve income, education, health care, environment, health and medical costs, genetic variables, etc. (Jakovljevic et al., 2016). Moreover, among all input factors, the health expenditure factors are the most commonly utilized variable, while measuring the health condition of a country (Jakovljevic et al., 2016; Kim & Lane, 2013). This study intends to draws it theoretical inspiration from the Grossman (1972) theory and adapt its framework to model the Nigeria's phenomenon.

3.3 Model Specification

Convention model specification procedures in economics generally involves three compartments which is the functional form of the model, the mathematical form and the econometric form respectively. The functional form (adapting from the theoretical framework of Grossman's health production model) represents the relationship between the dependent variable (life expectancy) and the key independent variables (health care financing, institutional quality) and control variables. This is expressed below thus:

$$LE=f\ (HCF, IQ, X) \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (1)$$

Where:

LE: Life Expectancy (dependent variable)

HCF: Health Care Financing (key independent variable i.e., capital health expenditure, recurrent health expenditure and out-of-pocket health expenditure)

IQ: Institutional Quality (key moderating/mediating variable i.e., composite index computed for institutional quality)

X: Vector of control variables (i.e, GDP per capita and Total Population).

To establish the explicit functional relationship, the model can be mathematically specified as:

$$LE_t = \alpha_0 + \alpha_1 HCF_t + \alpha_2 IQ_t + \alpha_3 (HCF * IQ)_t + \sum_{j=1}^{k} \alpha_j X_{tj} \ldots\ldots\ldots\ldots\ldots\ldots\ldots..(2)$$

Where:

t represents time period/time dimension (annual),

α_0 is the intercept term;

$\alpha_1, \alpha_2, \alpha_3$ are coefficients representing the effect of each variable;

$HCF * IQ$ denotes interaction term capturing the moderating effect of institutional quality on health care financing;

X_{tj} represents control variables influencing life expectancy.

The econometric model extends the mathematical representation to accommodate the stochastic term/error term. Thus, we have;

$$LE_t = \alpha_0 + \alpha_1 HCF_t + \alpha_2 IQ_t + \alpha_3 (HCF * IQ)_t + \sum_{j=1}^{k} \alpha_j X_{tj} + \varepsilon_t \ldots\ldots\ldots\ldots\ldots\ldots\ldots..(3)$$

Here: ε_t is the error term.

A-priori expectation

$\alpha_1, > 0$ = Health care financing positively impacts life expectancy.
$\alpha_2 > 0$ = Higher institutional quality positively influences life expectancy.
$\alpha_3 > 0$ = A positive and significant would imply that better institutional quality enhances the effectiveness of health care financing in improving life expectancy.

3.3.1 Estimation procedure

Four testing methodologies is employed to identify the study's objectives experimentally. This involves the pre-test (unit root and co-integration tests), formal analysis (Autoregressive Distributed Lagged (ARDL) technique) and the diagnostic tests (……..)

3.3.1.1 Unit Root Test

The study employs both the Phillip Peron (PP) and the Augmented Dickey-Fuller (ADF) tests for the unit roots in order to prevent spuriousness that could arise from the instability of certain variables in the analysis. The basis for these tests' argument is that big data sets, like the data set to be used for the study analysis, are best served by the ADF test, an improved Dickey-Fuller test. Conversely, the PP test is the most appropriate for handling basic econometric problems such as serial correlation and heteroscedasticity. The unit root test is conducted using a variety of equations, taking into account intercepts, trends, and the case when neither the intercepts nor the trends are included in the equations. The various considerations of the equations for both ADF and PP test take the following forms;

$$\Delta\gamma_t = \alpha_* + \alpha_{**}\gamma_{t-1} + \sum_{i=1}^{k} \beta_* \Delta\gamma_{t-1} + \eta_t \ldots\ldots\ldots\ldots (4)$$

$$\Delta\gamma_t = \alpha_* + \beta T + \alpha_{**}t + \alpha_{***}\gamma_{t-1} + \sum_{i=1}^{k} \beta_* \Delta\gamma_{t-1} + \eta_t \ldots\ldots\ldots\ldots.. (5)$$

$$\Delta\gamma_t = \beta T + \alpha_{***}\gamma_{t-1} + \sum_{i=1}^{k} \beta_* \Delta\gamma_{t-1} + \eta_t \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots.. (6)$$

$$\Delta\gamma_t = \alpha_* + \alpha_{**}T + \beta_* X_{t-1} + \eta_t \ldots\ldots\ldots\ldots\ldots\ldots.. (7)$$

$$\Delta\gamma_t = \alpha_{**}T + \beta_* X_{t-1} + \eta_t \ldots\ldots\ldots\ldots (8)$$

$$\Delta\gamma_t = \beta_* X_{t-1} + \eta_t \ldots\ldots\ldots\ldots\ldots (9)$$

The above equations (4) to (6) show the ADF test equations at constant, constant and trend, and without constant and trend respectively, where Δ represents the first difference operator; T is the time trend; and η is the error term; β_* and α_{*s} are parameters. On the other hand, the PP test with or without intercepts and trend are represented from equations (7) to (9) above.

3.3.1.2 Co-integration Test

The co-integration test is required to verify that the model's mixed series will eventually converge. This is because a series' inability to stabilize over the short term does not imply that it will eventually converge with other model series. The ARDL bounds test is used for this investigation. The F-statistic of the limits test must be higher than the upper bound at any selected significance level in order to confirm the probability of a long run. If not, the series do

not converge and only short-term relationships are present. On the other hand, the poor convergence, the F-statistic falling between the boundaries, and the lack of a lengthy run make this impossible to determine. The relative stationarity of the series is essentially what determines which ARDL technique to use, aside from the specifics of this investigation. Models with series of mixed order of integration—that is, where at least one variable in the model is not stationary until it is differentiated once—can be accommodated by the ARDL technique.

3.2. Autoregressive Distributed Lagged (ARDL) Model

Furthermore, the presentation of the ARDL model specification is shown below;

$$\Delta Y_t = \rho_0 + \pi T + \sum_{i=1}^{pq} \varphi_i \Delta Y_{t-i} + \sum_{j=1}^{mn} \varphi_j \Delta X_{t-j} + \delta_1 Y_{t-1} + \delta_2 X_{t-1} + \mu_t \ldots\ldots\ldots\ldots (10)$$

Where Y represents the dependent variables as defined above, X represent the regressor. Δ represents the first difference operator, ρ_0 is the drift, μ_t is the error term, T is the trend of the model and i=1…pq lag periods. In the ARDL model, an error correction mechanism existed by which changes in the dependent variables were modeled as a function of the level of the disequilibrium in the cointegrating relationship, captured by the error-correction term (ECT), as well as changes in the other explanatory variables to capture all short-term relations among variables.

Accordingly, equation (10) is further re-specified to capture the error correction mechanism as developed by Pesaran et al. (2001). Thus, we have;

$$\Delta Y_t = \rho_0 + \pi T + \sum_{i=1}^{pq} \varphi_i \Delta Y_{t-i} + \sum_{j=1}^{mn} \varphi_j \Delta X_{t-j} + \delta_1 Y_{t-1} + \delta_2 X_{t-1} + \Omega \boldsymbol{ECM_{t-i}} + \mu_t \ldots\ldots\ldots\ldots (11)$$

Here, $\boldsymbol{ECM_{t-i}}$ is the lagged error correction term calculated as $\boldsymbol{ECM_{t-i}} = \boldsymbol{Y_{t-1}} - \hat{\rho} - \widehat{\delta_2}\boldsymbol{X_{t-1}}$

4. Discussion, Conclusion and Recommendations

This chapter is dedicated to the presentation of the study's results as well as a discussion of the findings. It is divided into two sections: preliminary results and the main findings. The main results were evaluated using the autoregressive distributed lag (ARDL) model to answer the study's research questions and test the hypothesis raised in the preceding chapter. However,

we set the tone of this chapter by recapping the ARDL model specified in the previous chapter for an easy progression and understanding of what to expect in this chapter.

4.1 Data Presentation

YEAR	CHE	RHE	OUE	LEX	POP	GDP	C_IQ	CHEIQ	RHEIQ	OUEIQ
1984	18.06819	8.830604	11.20056	46.55	82526443	164.21	3.486111	62.98771	30.78447	39.04639
1985	17.92801	11.80473	11.20056	46.317	84897973	185.98	3.583333	64.24203	42.30028	40.13534
1986	17.88273	11.80473	11.20056	45.975	87235953	196.17	3.3125	59.23654	39.10317	37.10185
1987	17.82091	9.185776	11.20056	46.018	89591716	242.26	3.548611	63.23947	32.59675	39.74643
1988	17.95964	7.716393	11.20056	46.072	92020090	312.50	3.701389	66.47562	28.56137	41.45762
1989	17.87721	4.742267	11.13259	46.182	94531014	410.77	3.951389	70.63983	18.73854	43.9892
1990	17.78521	4.742267	10.93901	46.037	97120925	489.77	4.027778	71.63486	19.1008	44.05989
1991	17.9872	7.716393	10.74542	45.691	99720162	584.25	4.076389	73.32284	31.45502	43.80251
1992	18.06819	10.33535	10.68777	45.668	1.02E+08	897.12	4.458333	80.554	46.07842	47.64962
1993	18.06819	15.28035	10.88135	45.788	1.05E+08	1,244.80	4.423611	79.92663	67.59432	48.13487
1994	17.84558	24.25014	11.20056	45.513	1.08E+08	1,751.28	4.527778	80.80083	109.7992	50.71364
1995	17.8003	21.27601	11.38383	45.487	1.11E+08	3,069.43	4.361111	77.62909	92.78704	49.64616
1996	17.93903	18.65706	11.31587	45.567	1.14E+08	4,045.32	4.423611	79.35531	82.53156	50.057
1997	17.95964	16.70198	10.99666	45.792	1.17E+08	4,374.50	3.930556	70.59137	65.64807	43.22299
1998	18.00492	4.74	10.81339	46.036	1.2E+08	4,756.71	3.541667	63.76743	16.79553	38.29741
1999	17.97329	16.64	12.69434	46.614	1.23E+08	5,426.47	4.180556	75.13834	69.55931	53.0694
2000	17.65199	15.22	10.6198	47.193	1.26E+08	6,990.62	3.8125	67.2982	58.01894	40.48798
2001	18.48872	24.52	11.23149	47.619	1.3E+08	8,150.02	3.885417	71.83637	95.27925	43.63903
2002	18.06385	40.62	11.75038	47.928	1.33E+08	11,383.66	3.232639	58.39392	131.3144	37.98475
2003	39.07724	33.27	28.45385	48.441	1.37E+08	13418.012898+G418328	3.461806	135.2778	115.1673	98.50168
2004	44.16435	34.20	28.50715	48.767	1.41E+08	17,938.38	3.642361	160.8625	124.5688	103.8333
2005	53.92662	55.70	35.57567	49.297	1.45E+08	22,884.90	3.684028	198.6672	205.2003	131.0617

2006	65.72379	62.25	46.30819	49.73	1.49E+08	30,063.96	3.833333	251.9412	238.6389	177.5147
2007	69.15039	81.91	49.05359	50.033	1.53E+08	34,318.67	3.652778	252.591	299.1967	179.1818
2008	80.09986	98.22	58.2782	50.225	1.58E+08	39,542.43	3.625	290.362	356.045	211.2585
2009	68.00686	90.20	50.64807	50.712	1.62E+08	43,012.51	3.802083	258.5677	342.9479	192.5682
2010	75.58925	99.10	58.11112	50.945	1.67E+08	54,612.26	3.774306	285.2969	374.0337	219.3291
2011	83.10648	231.80	62.10144	51.357	1.71E+08	62,980.40	3.833333	318.5748	888.5667	238.0555
2012	91.06048	197.90	66.3324	51.497	1.76E+08	71,713.94	3.75	341.4768	742.125	248.7465
2013	100.817	180.00	71.50621	51.707	1.81E+08	80,092.56	3.701389	373.163	666.25	264.6723
2014	106.1196	195.98	76.25141	51.791	1.86E+08	89,043.62	3.565972	378.4197	698.8477	271.9104
2015	96.28329	257.70	69.21813	51.841	1.91E+08	94,144.96	3.6875	355.0446	950.2688	255.2419
2016	78.23602	200.82	58.82334	52.043	1.95E+08	101,489.49	3.673611	287.4087	737.7492	216.0941
2017	72.77431	245.19	56.31849	52.305	2E+08	113,711.63	3.684028	268.1026	903.2795	207.4789
2018	65.70272	296.44	49.89845	52.554	2.05E+08	127,736.83	3.826389	251.4042	1134.302	190.9309
2019	69.68148	388.23	49.82019	52.91	2.09E+08	144,210.49	3.982639	277.5162	1546.19	198.4158
2020	69.76185	423.36	52.0952	52.887	2.14E+08	152,324.07	3.923611	273.7184	1661.097	204.4013
2021	83.84061	386.24	63.9229	52.676	2.19E+08	173,527.66	3.909722	327.7935	1510.108	249.9208
2022	68.38202	437.52	69.38215	53.633	2.23E+08	199,336.04	3.958333	270.6788	1731.855	274.6377
2023	74.42798	468.64	76.43351	53.81133	2.28E+08	229,912.94	3.965278	295.1276	1858.283	303.0801

Table 4.1: *Data Presentation*

Source: *Author's Computation 2025*

4.2 Discussions of Findings

4.2.1 Descriptive Analysis

Conventional practice in the literature is to demonstrate the characteristics of data. This study followed these procedures, presenting the descriptive statistics and trend analysis in the appendix, respectively. The results presented in Table 4.2 in the Appendix show that the capital health expenditure (CHE) has an annual average of ₦48.43 billion, with significant volatility

due to a high standard deviation of ₦30.80 billion while peaking at ₦106.12 billion in the year 2014. The dataset exhibits positive skewness of 0.308 and platykurtic distribution with kurtosis less than 3, indicating occasional high-spending years, likely linked to infrastructure investments or disease outbreaks (). However, Recurrent Health Expenditure (RHE) shows even greater fluctuations, with an annual average of ₦118.49 billion but a median of just ₦37.41 billion, reflecting sporadic surges while reaching its maximum in 2023, amounting to ₦468.64 billion due to health crises or policy shifts (IMF, 2023;). The high skewness of 1.181 and mesokurtic distribution of kurtosis approximately to 3 confirm a non-normal distribution, further validated by the significant Jarque-Bera test.

Out-of-Pocket expenditure (OUE) indicated an average of ₦34.74 billion, with extreme values of ₦10.62 billion and ₦76.43 billion for minimum and maximum in years 2000 and 2023, respectively, suggesting varying household healthcare burdens. The right-skewed distribution of 0.298 skewness and platykurtic distribution due to a kurtosis of 1.419 less than 3 implies periods of heightened financial strain, possibly during economic downturns or reduced public health funding (WHO, 2021). Institutional quality (C_IQ), on the other hand, exhibits a relatively stable annual average of 3.84 index and a standard deviation of 0.30, though slightly right-skewed with a value of 0.572 and a mesokurtic distribution with its kurtosis approximately 3, hinting at intermittent governance improvements.

Considering the control variables, the gross domestic product (GDP) exhibits stark disparities; the mean of ₦48,767.29 billion is triple the median of ₦15,678.20 billion, with a maximum of ₦229,912.9 billion in the year 2023 and a minimum of ₦164.21 billion in the year 1984, indicating oil-driven booms. The high standard deviation of 62,694.02 and skewness of 1.310 while having a leptokurtic distribution due to its kurtosis higher than 3 underscore Nigeria's growth volatility, consistent with resource-dependent economies (Acemoglu & Robinson, 2012). However, the GDP's skewness implies that health budgets may shrink during oil slumps while diversifying revenue sources is critical to safeguard health spending (UNDP, 2022). Thus, the Jarque-Bera test of significant p-value rejects normality, aligning with GDP's erratic trends. In addition, population (POP) revealed an annual average of 146 million, with moderate dispersion of a standard deviation of 44.56 million, reflecting a steady growth in Nigeria. POP exhibits relatively symmetric skewness less than 0.6 with normal-like kurtosis, and their Jarque-Bera tests greater than the significant p-value of 0.05 suggest normality.

4.2.2 Unit Root: Augmented Dicker-Fuller (ADF)

Ascertaining whether the series of variables in this study reverts to their means to confirm their stationarity is crucial for two reasons. First, it helps prevent reporting spurious results. Second, it clarifies whether the ARDL model in this study is justified. Augmented Dickey-Fuller was used to perform this task in order to provide robustness to the stationarity of each variable.

The results, presented in Table 4.3 of the Appendix, reveal that the unit root test exhibits mixed significant results while all the variables are in log form except life expectancy. Aside from life expectancy (LEX), the log of recurrent health expenditure (LNRHE), and the log of population (LNPOP), which have their unit root in their series at levels, other variables show properties of a unit root after the first difference. This implies that the LEX, LNRHE, and LNPOP series revert to their mean values after deviating at their level form with their t-statistic values of -4.6561, -3.2194, and -4.4800 higher than their critical values, which indicates that they were stationary at levels of 1%, 10%, and 1%, respectively. However, the other series exhibited the same mean-reverting properties at the first difference. That is, there is stationarity in the series of the log of capital health expenditure (LNCHE) with a t-stat of -5.4347, the log of out-of-pocket (LNOUE) with a t-stat of -5.7637, the log of institutional quality (LNC_IQ) with a t-stat of -7.8727, and the log of gross domestic product (LNGDP) with a t-stat of -4.2145, by differencing their series once, which is stationary at 1%, 1%, 1%, and 5%, respectively.

The unit root test results show that the stationarity of the variables in this study is mixed between the levels (I(0)) and first difference (I(1)). This further strengthens the choice of employing ARDL as an estimation approach in this study because it can handle the stationarity of mixed orders, with the exception of the second difference (Salisu and Isah, 2017). In addition, the dependent variable, life expectancy, and the independent variables, such as current health expenditure, out-of-pocket expenditure, and institutional quality, are stationary at the first difference, and at least one control variable is stationary at levels. This conforms to the specifications of the bound test and makes ARDL an appropriate technique for this study.

4.2.3 Main Findings

This section presents and discusses the results of this study, in tandem with its objectives. Consequently, this section is divided into two parts, with objective three answered in each part.

4.2.3.1 Interactive Role of Institutional Quality on CHE-Life Expectancy Nexus

The analysis of capital health expenditure (CHE) reveals a complex relationship with life expectancy that is significantly mediated by institutional quality. In the baseline model without institutional quality controls, CHE demonstrates a strong positive and statistically significant impact on life expectancy (LEX), with a coefficient of 2.647 ($p = 0.019$). This suggests that a 1% increase in capital health expenditure leads to a 2.65-year improvement in life expectancy, supporting existing literature that links infrastructure investments in healthcare to improved population health outcomes (Gallet & Doucouliagos, 2017). These capital expenditures, which typically include investments in medical facilities, equipment, and health infrastructure, appear to create the physical foundation for better healthcare delivery. However, when institutional quality is introduced into the model, the picture changes dramatically. The interaction term between CHE and institutional quality (LNCHE_IQ) becomes negative and statistically insignificant (coefficient = -0.307, $p = 0.886$), while institutional quality alone shows a significant negative effect (coefficient = -29.239, $p = 0.025$). This striking finding suggests that weak governance structures effectively undermine the potential benefits of capital health investments, likely through mechanisms such as corruption, misallocation of resources, or inefficient implementation of health projects (Murshed & Ahmed, 2018). The results align with numerous African studies demonstrating how institutional weaknesses can negate the positive impacts of health sector investments (Okunade, et al., 2025). In practical terms, this means that while Nigeria may allocate substantial funds to build hospitals and purchase medical equipment, these investments may fail to translate into actual health improvements if they are not accompanied by strong governance frameworks, proper oversight mechanisms, and effective management systems. The policy implications are clear: improving institutional quality must be a parallel priority alongside increasing health sector investments to ensure that capital expenditures actually achieve their intended outcomes in terms of population health.

4.2.3.2 Interactive Role of Institutional Quality on RHE-Life Expectancy Nexus

The examination of recurrent health expenditure (RHE) presents a different pattern from capital expenditures in its relationship with life expectancy. In the baseline model, RHE shows a positive but statistically insignificant effect on LEX (coefficient = 3.171, $p = 0.713$). This suggests that routine health spending on items like salaries, drugs, and maintenance may not independently drive significant improvements in life expectancy without complementary factors or investments. The relative ineffectiveness of recurrent expenditures could stem from

various systemic issues common in developing country health systems, including leakage of funds, inefficient allocation, or poor quality of services delivered (WHO, 2020). When institutional quality is introduced to the model, the interaction term (LNRHE_IQ) remains positive but still insignificant (coefficient = 0.740, p = 0.109), while institutional quality itself maintains a significant negative coefficient (coefficient = -17.440, p = 0.024). These findings mirror documented challenges in Nigeria and similar studies where recurrent health budgets often fail to translate into quality service delivery due to governance weaknesses like ghost workers, payroll fraud, and procurement irregularities (Olanrawaju, 2019). The persistent negative effect of institutional quality indicates that poor governance may be particularly detrimental to the effective utilization of recurrent health funds, which require consistent, well-managed systems to produce results. Unlike capital expenditures that create physical assets, recurrent spending depends entirely on ongoing institutional capacity to deliver services efficiently. The results suggest that simply increasing health sector salaries or drug budgets without addressing fundamental governance issues is unlikely to yield significant improvements in population health outcomes. This has important implications for health financing strategies, emphasizing the need to combine recurrent budget increases with comprehensive institutional reforms that improve accountability, transparency, and management effectiveness throughout the health system.

4.2.3.3 Interactive Role of Institutional Quality on OUE-Life Expectancy Nexus

The analysis of out-of-pocket health expenditures (OUE) reveals yet another dimension of the health financing-life expectancy relationship. In the baseline model, OUE shows a positive but statistically insignificant association with LEX (coefficient = 1.704, p = 0.128), suggesting that household health spending alone does not reliably contribute to improved life expectancy at the population level. This finding aligns with extensive literature documenting how out-of-pocket payments for healthcare often create financial barriers to access, particularly for poorer households, and can lead to delayed care-seeking or foregoing treatment altogether (Onah & Govender, 2014). When institutional quality is added to the model, the interaction term (LNOUE_IQ) remains positive and insignificant (coefficient = 0.174, p = 0.893), while institutional quality itself shows a negative but also insignificant coefficient (coefficient = -24.140, p = 0.252). The consistently insignificant results across both models indicate that out-of-pocket expenditures may be too variable and inequitable in their effects to reliably improve population health outcomes, regardless of institutional context. This reinforces the argument that reliance on household spending for healthcare is an inadequate strategy for improving

national health indicators, as it disproportionately burdens lower-income groups while failing to generate consistent health gains (Musango, et al., 2012). The findings strongly support the case for reducing dependence on out-of-pocket payments through alternative financing mechanisms like health insurance or tax-based funding. Particularly in weak institutional environments where regulatory oversight is limited, out-of-pocket systems tend to exacerbate health inequalities without delivering commensurate improvements in overall population health. The policy implications point toward the urgent need to implement protective financing mechanisms like Nigeria's National Health Insurance Scheme (NHIS) on a much broader scale, combined with efforts to improve the quality and accessibility of services that households are paying for. Only by addressing both the financial and quality dimensions of healthcare simultaneously can meaningful improvements in life expectancy be achieved.

4.3 Conclusion

This study examined the relationship between health expenditures (capital, recurrent, and out-of-pocket) and life expectancy in Nigeria, with institutional quality as a moderating factor. The findings reveal that while capital health expenditure (CHE) significantly improves life expectancy when analyzed independently, its benefits are nullified by poor institutional quality, highlighting governance deficiencies such as corruption and inefficiency. However, recurrent health expenditure (RHE) showed no significant impact on life expectancy, underscoring systemic issues like fund leakage and mismanagement. In tandem, out-of-pocket expenditure (OUE) also failed to yield meaningful improvements, exacerbating inequities without enhancing population health outcomes. These results emphasize that Nigeria's health financing strategies, whether capital investments, recurrent spending, or household payments are ineffective without robust institutional frameworks. The study aligns with global evidence that health investments require strong governance to translate into tangible health improvements.

4.4 Recommendations

To enhance the effectiveness of health expenditures, the following measures are recommended:

1. Strengthen Institutional Quality: Implementing more rigorous anti-corruption measures, transparent budgeting, and performance-based financing to ensure efficient use of allocated health funds.

2. Reform Recurrent Spending: Tie recurrent health expenditure allocations to measurable outcomes such as service delivery metrics and eliminate ghost workers through biometric verification systems in Nigeria's health sector.

3. Minimize Reliance on OUE: Through the expansion of the National Health Insurance Scheme (NHIS) to cover more Nigerians, particularly low-income populations, and subsidize essential healthcare services.

4. Adopting Integrated Policies: Synergizing health financing reforms with institutional capacity-building programs to ensure sustainable improvements in life expectancy in Nigeria.

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APPENDIX

Table 4.2: Descriptive Statistic

	LEX	CHE	RHE	OUE	C_IQ	POP	GDP
Mean	48.93023	47.42763	118.4866	34.73587	3.835069	1.46E+08	48767.29
Median	48.60400	41.62080	37.41071	28.48050	3.807292	1.39E+08	15678.20
Maximum	53.81133	106.1196	468.6388	76.43351	4.527778	2.28E+08	229912.9
Minimum	45.48700	17.65199	4.742267	10.61980	3.232639	82526443	164.2100
Std. Dev.	2.876405	30.79659	142.8669	24.76915	0.303922	44562071	62694.02
Skewness	0.204856	0.307597	1.180811	0.298425	0.572311	0.305722	1.310292
Kurtosis	1.463239	1.519117	3.083382	1.418688	3.049419	1.827533	3.733344
Jarque-Bera	4.215832	4.285798	9.307019	4.761297	2.187668	2.914238	12.34209
Probability	0.121491	0.117314	0.009528	0.092491	0.334930	0.232906	0.002089
Sum	1957.209	1897.105	4739.464	1389.435	153.4028	5.83E+09	1950692.
Sum Sq. Dev.	322.6746	36988.78	796026.5	23926.92	3.602382	7.74E+16	1.53E+11
Observations	40	40	40	40	40	40	40

Source: *E-views 9*

Table 4.3: Unit Root Results (Augment Dickey-Fuller Test)

LEX (@Level)

Null Hypothesis: LEX has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 6 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-4.656149	0.0038
Test critical values:	1% level	-4.262735	
	5% level	-3.552973	
	10% level	-3.209642	

*MacKinnon (1996) one-sided p-values.

LNCHE (@1st Diff)

Null Hypothesis: D(LNCHE) has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-5.434738	0.0004
Test critical values:	1% level	-4.219126	
	5% level	-3.533083	
	10% level	-3.198312	

*MacKinnon (1996) one-sided p-values.

LNRHE (@Level)

Null Hypothesis: LNRHE has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-3.219412	0.0956
Test critical values:	1% level	-4.211868	
	5% level	-3.529758	
	10% level	-3.196411	

*MacKinnon (1996) one-sided p-values.

LNOUE (@1st Diff)

Null Hypothesis: D(LNOUE) has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-5.763741	0.0002
Test critical values:	1% level	-4.219126	
	5% level	-3.533083	
	10% level	-3.198312	

*MacKinnon (1996) one-sided p-values.

LNC_IQ (@ 1st Diff)

Null Hypothesis: D(LNC_IQ) has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-7.872720	0.0000
Test critical values:	1% level	-4.219126	
	5% level	-3.533083	
	10% level	-3.198312	

*MacKinnon (1996) one-sided p-values.

LNPOP (@ Level)

Null Hypothesis: LNPOP has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 3 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-4.480035	0.0054
Test critical values:	1% level	-4.234972	
	5% level	-3.540328	
	10% level	-3.202445	

*MacKinnon (1996) one-sided p-values.

LNGDP (@1st Diff)

Null Hypothesis: D(LNGDP) has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-4.214454	0.0101
Test critical values:	1% level	-4.219126	
	5% level	-3.533083	
	10% level	-3.198312	

*MacKinnon (1996) one-sided p-values.

LNCHE*IQ (@ 1st Diff)

Null Hypothesis: D(LNCHE_IQ) has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-6.331544	0.0000
Test critical values:	1% level	-4.219126	
	5% level	-3.533083	
	10% level	-3.198312	

*MacKinnon (1996) one-sided p-values.

LNRHE*IQ (@ Level)

Null Hypothesis: LNRHE_IQ has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-3.349882	0.0733
Test critical values:	1% level	-4.211868	
	5% level	-3.529758	
	10% level	-3.196411	

*MacKinnon (1996) one-sided p-values.

LNOUE*IQ (@ 1st Diff)

Null Hypothesis: D(LNOUE_IQ) has a unit root
ysisCExogenous: Constant, Linear Trend
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-6.858771	0.0000
Test critical values:	1% level	-4.219126	
	5% level	-3.533083	
	10% level	-3.198312	

*MacKinnon (1996) one-sided p-values.

Source: *E-views 9*

ARDL ANALYSIS WITHOUT COMPOSITE INSTITUTIONAL QUALITY

Model 1: LEX = f(LNCHE, LNPOP, LNGDP)

ARDL Cointegrating And Long Run Form
Dependent Variable: LEX
Selected Model: ARDL(4, 4, 4, 4)
Date: 04/10/25 Time: 20:54
Sample: 1984 2023
Included observations: 36

Cointegrating Form

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(LEX(-1))	-0.725616	0.326402	-2.223078	0.0410
D(LEX(-2))	-0.447775	0.263915	-1.696666	0.1091
D(LEX(-3))	-0.463460	0.282982	-1.637771	0.1210
D(LNCHE)	-0.492448	0.362560	-1.358251	0.1932
D(LNCHE(-1))	0.230650	0.326432	0.706579	0.4900
D(LNCHE(-2))	0.006962	0.327097	0.021284	0.9833
D(LNCHE(-3))	0.481780	0.290631	1.657703	0.1169
D(LNPOP)	132.825044	166.131833	0.799516	0.4357
D(LNPOP(-1))	-706.504510	519.017714	-1.361234	0.1923
D(LNPOP(-2))	785.637733	415.186549	1.892252	0.0767
D(LNPOP(-3))	-352.799711	157.454394	-2.240647	0.0396
D(LNGDP)	-0.288650	0.458969	-0.628910	0.5383
D(LNGDP(-1))	1.466646	0.625932	2.343138	0.0324
D(LNGDP(-2))	-0.946114	0.683418	-1.384386	0.1852
D(LNGDP(-3))	-0.883925	0.549036	-1.609958	0.1270
CointEq(-1)	0.369928	0.258222	1.432595	0.1712

Cointeq = LEX - (2.6469*LNCHE + 28.6589*LNPOP -3.5217*LNGDP -472.9356)

Long Run Coefficients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LNCHE	2.646938	1.013803	2.610900	0.0189
LNPOP	28.658925	11.637201	2.462699	0.0255
LNGDP	-3.521729	1.463935	-2.405659	0.0286
C	-472.935639	211.659297	-2.234419	0.0401

Source: *E-views 9*

Model 2: LEX = f(LNRHE, LNPOP, LNGDP)

ARDL Cointegrating And Long Run Form
Dependent Variable: LEX
Selected Model: ARDL(2, 0, 4, 4)
Date: 04/10/25 Time: 21:01
Sample: 1984 2023
Included observations: 36

Cointegrating Form

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(LEX(-1))	-0.394107	0.225680	-1.746307	0.0947
D(LNRHE)	-0.151593	0.140070	-1.082263	0.2909
D(LNPOP)	168.394930	166.044638	1.014155	0.3215
D(LNPOP(-1))	-612.011931	510.170549	-1.199622	0.2430
D(LNPOP(-2))	522.544880	389.273822	1.342358	0.1932
D(LNPOP(-3))	-234.339971	139.548614	-1.679271	0.1072
D(LNGDP)	-0.030618	0.486752	-0.062904	0.9504
D(LNGDP(-1))	1.536976	0.605348	2.538995	0.0187
D(LNGDP(-2))	-0.335103	0.683402	-0.490346	0.6287
D(LNGDP(-3))	-1.023550	0.530205	-1.930478	0.0665
CointEq(-1)	0.047799	0.129903	0.367959	0.7164

Cointeq = LEX - (3.1715*LNRHE + 123.6849*LNPOP -17.4917*LNGDP -2192.1706)

Long Run Coefficients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LNRHE	3.171461	8.522747	0.372117	0.7134
LNPOP	123.684941	247.050140	0.500647	0.6216
LNGDP	-17.491739	39.064494	-0.447766	0.6587
C	-2192.170609	4457.197373	-0.491827	0.6277

Source: *E-views 9*

Model 3: LEX = f(LNOUE, LNPOP, LNGDP)

ARDL Cointegrating And Long Run Form
Dependent Variable: LEX
Selected Model: ARDL(1, 0, 4, 4)
Date: 04/10/25 Time: 21:04
Sample: 1984 2023
Included observations: 36

Cointegrating Form

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(LNOUE)	0.331110	0.233669	1.417005	0.1699
D(LNPOP)	151.151053	170.373308	0.887176	0.3842
D(LNPOP(-1))	-663.034606	511.610817	-1.295975	0.2078
D(LNPOP(-2))	576.617891	390.897874	1.475111	0.1537
D(LNPOP(-3))	-240.748736	138.106128	-1.743215	0.0947
D(LNGDP)	-0.328353	0.449269	-0.730860	0.4722

D(LNGDP(-1))	1.381532	0.590469	2.339722	0.0283
D(LNGDP(-2))	-0.625148	0.637714	-0.980296	0.3371
D(LNGDP(-3))	-0.767926	0.482701	-1.590894	0.1253
CointEq(-1)	-0.194364	0.144907	-1.341305	0.1929

Cointeq = LEX - (1.7036*LNOUE -11.2924*LNPOP + 2.0453*LNGDP + 250.6717)

Long Run Coefficients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LNOUE	1.703557	1.077602	1.580878	0.1276
LNPOP	-11.292362	28.225857	-0.400072	0.6928
LNGDP	2.045346	3.241985	0.630893	0.5343
C	250.671708	514.030146	0.487660	0.6304

Source: *E-views 9*

ARDL ANALYSIS WITH PRESENCE OF COMPOSITE INSTITUTIONAL QUALITY

Model 1: LEX = f(LNCHE*C_IQ, C_IQ, LNPOP, LNGDP)

ARDL Cointegrating And Long Run Form
Dependent Variable: LEX
Selected Model: ARDL(4, 0, 4, 4, 4)
Date: 04/10/25 Time: 21:07
Sample: 1984 2023
Included observations: 36

Cointegrating Form

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(LEX(-1))	-0.644779	0.192995	-3.340909	0.0045
D(LEX(-2))	-0.566473	0.203049	-2.789837	0.0137
D(LEX(-3))	-0.509129	0.211740	-2.404506	0.0296
D(LNCHE_IQ)	-0.051643	0.310944	-0.166083	0.8703
D(LNC_IQ)	-0.908593	0.906050	-1.002807	0.3319
D(LNC_IQ(-1))	1.165231	0.618322	1.884505	0.0790
D(LNC_IQ(-2))	0.941613	0.663055	1.420113	0.1760
D(LNC_IQ(-3))	1.340165	0.699187	1.916747	0.0745
D(LNPOP)	192.448323	128.924966	1.492716	0.1562
D(LNPOP(-1))	-883.768410	402.690125	-2.194661	0.0443
D(LNPOP(-2))	1066.317689	342.948639	3.109264	0.0072
D(LNPOP(-3))	-530.405332	137.714081	-3.851497	0.0016
D(LNGDP)	-0.220424	0.382320	-0.576545	0.5728
D(LNGDP(-1))	1.455537	0.572414	2.542807	0.0225
D(LNGDP(-2))	-0.577085	0.512105	-1.126888	0.2775
D(LNGDP(-3))	-0.806727	0.468095	-1.723424	0.1054
CointEq(-1)	-0.168224	0.178855	-0.940557	0.3618

Cointeq = LEX - (-0.3070*LNCHE_IQ -29.2390*LNC_IQ -35.3571*LNPOP + 6.2947*LNGDP + 723.9570)

Long Run Coefficients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LNCHE_IQ	-0.306988	2.098228	-0.146308	0.8856
LNC_IQ	-29.238955	28.093894	-1.040758	0.3145
LNPOP	-35.357063	59.367126	-0.595566	0.5603
LNGDP	6.294722	8.554544	0.735834	0.4732
C	723.956995	1113.003888	0.650453	0.5252

Source: *E-views 9*

Model 2: LEX = f(LNRHE*C_IQ, C_IQ, LNPOP, LNGDP)

ARDL Cointegrating And Long Run Form
Dependent Variable: LEX
Selected Model: ARDL(4, 1, 4, 4, 4)
Date: 04/10/25 Time: 21:15
Sample: 1984 2023
Included observations: 36

Cointegrating Form

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(LEX(-1))	-0.528385	0.193385	-2.732295	0.0162
D(LEX(-2))	-0.578037	0.210051	-2.751884	0.0156
D(LEX(-3))	-0.494123	0.202489	-2.440250	0.0286
D(LNRHE_IQ)	0.104230	0.127445	0.817840	0.4271
D(LNC_IQ)	-0.840884	0.705468	-1.191952	0.2531
D(LNC_IQ(-1))	1.438525	0.638960	2.251354	0.0409
D(LNC_IQ(-2))	1.191305	0.706691	1.685751	0.1140
D(LNC_IQ(-3))	1.681717	0.696942	2.412994	0.0301
D(LNPOP)	205.543713	124.753099	1.647604	0.1217
D(LNPOP(-1))	-870.254203	393.072370	-2.213980	0.0439
D(LNPOP(-2))	1129.777838	329.838980	3.425241	0.0041
D(LNPOP(-3))	-580.194878	132.684012	-4.372757	0.0006
D(LNGDP)	-0.674776	0.494300	-1.365113	0.1938
D(LNGDP(-1))	1.329849	0.556102	2.391374	0.0314
D(LNGDP(-2))	-0.945116	0.572661	-1.650393	0.1211
D(LNGDP(-3))	-0.590094	0.456920	-1.291459	0.2175
CointEq(-1)	-0.339465	0.165351	-2.052993	0.0593

Cointeq = LEX - (0.7404*LNRHE_IQ -17.4404*LNC_IQ -12.9058*LNPOP + 2.5340*LNGDP + 301.6648)

Long Run Coefficients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LNRHE_IQ	0.740393	0.432973	1.710021	0.1093
LNC_IQ	-17.440412	6.912935	-2.522866	0.0244
LNPOP	-12.905767	17.891669	-0.721328	0.4826
LNGDP	2.533972	2.307040	1.098365	0.2906
C	301.664761	333.542312	0.904427	0.3811

Source: *E-views 9*

Model 3: LEX = f(LNOUE*C_IQ, C_IQ, LNPOP, LNGDP)

ARDL Cointegrating And Long Run Form
Dependent Variable: LEX
Selected Model: ARDL(4, 0, 4, 4, 4)
Date: 04/10/25 Time: 21:17
Sample: 1984 2023
Included observations: 36

Cointegrating Form

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(LEX(-1))	-0.620581	0.206946	-2.998754	0.0090
D(LEX(-2))	-0.546608	0.223933	-2.440942	0.0275
D(LEX(-3))	-0.520717	0.204335	-2.548350	0.0223
D(LNOUE_IQ)	0.036230	0.289224	0.125265	0.9020
D(LNC_IQ)	-1.104246	0.860823	-1.282780	0.2190
D(LNC_IQ(-1))	1.143772	0.613281	1.865005	0.0819
D(LNC_IQ(-2))	0.893494	0.646741	1.381534	0.1873
D(LNC_IQ(-3))	1.406795	0.665255	2.114671	0.0516
D(LNPOP)	183.584897	132.767432	1.382755	0.1870
D(LNPOP(-1))	-890.138593	402.243250	-2.212936	0.0428
D(LNPOP(-2))	1083.023454	336.799703	3.215631	0.0058
D(LNPOP(-3))	-538.322967	131.716795	-4.086973	0.0010
D(LNGDP)	-0.248168	0.388121	-0.639409	0.5322
D(LNGDP(-1))	1.519120	0.547482	2.774740	0.0142
D(LNGDP(-2))	-0.576643	0.512613	-1.124911	0.2783
D(LNGDP(-3))	-0.866264	0.455496	-1.901802	0.0766
CointEq(-1)	-0.207825	0.188524	-1.102378	0.2877

Cointeq = LEX - (0.1743*LNOUE_IQ -24.1400*LNC_IQ -26.8250*LNPOP + 4.9210*LNGDP + 562.6645)

Long Run Coefficients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LNOUE_IQ	0.174328	1.275615	0.136662	0.8931
LNC_IQ	-24.140006	20.279523	-1.190364	0.2524
LNPOP	-26.825006	43.253011	-0.620188	0.5444
LNGDP	4.920985	6.143341	0.801027	0.4356
C	562.664481	810.376868	0.694324	0.4981

Source: *E-views 9*